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A Blockchain-Assisted Federated Learning Framework for Secure and Self-Optimizing Digital Twins in Industrial IoT

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Abstract: Optimizing digital twins in the Industrial Internet of Things (IIoT) requires secure and adaptable AI models. The IIoT enables digital twins, virtual replicas of physical assets, to improve real-time decision-making, but challenges remain in trust, data security, and model accuracy. This paper presents a novel framework combining blockchain technology and federated learning (FL) to address these issues. By deploying AI models on edge devices and using FL, data privacy is maintained while enabling collaboration across industrial assets. Blockchain ensures secure data management and transparency, while explainable AI (XAI) enhances interpretability. The framework improves transparency, control, security, privacy, and scalability for self-optimizing digital twins in IIoT. A real-world evaluation demonstrates the framework's effectiveness in enhancing security, explainability, and optimization, offering improved efficiency and reliability for industrial operations.

Keywords: blockchain; digital twins; federated learning; Industrial Internet of Things



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1. Introduction

The Industrial Internet of Things (IIoT) represents a transformative shift in industrial operations, enabling enhanced connectivity and automation across diverse applications [1]. Central to IIoT are digital twins—virtual replicas of physical assets—that facilitate real-time monitoring, predictive maintenance, and operational optimization [2]. Digital twins bridge the physical and digital realms through synchronized data flows, creating opportunities for smarter decision-making, reduced downtime, and increased productivity [3]. The three integral components of digital twins—physical assets, their virtual counterparts, and the IIoT-enabled linkage between them—serve as the foundation for IIoT systems [4].

Incorporating AI-driven analytics into digital twins has expanded their capabilities but also introduced challenges related to data security, privacy, and the explainability of automated decisions [5]. Addressing these challenges necessitates a robust framework that integrates federated learning (FL), blockchain technology, and Explainable AI (XAI) to create a secure, scalable, and interpretable digital twin ecosystem.

FL has emerged as a promising approach for distributed machine learning, enabling the training of models directly on edge devices where data are generated [6]. This decentralized paradigm enhances privacy by eliminating the need for raw data to leave local devices

while also improving model adaptability by leveraging diverse datasets [7]. Figure 1 illustrates the distributed training concept within an IIoT context, demonstrating how edge devices contribute to a global model by processing localized data securely. While this figure focuses on autonomous vehicles, the underlying principles apply broadly across the IIoT spectrum, encompassing diverse sectors, such as manufacturing, energy, and healthcare.

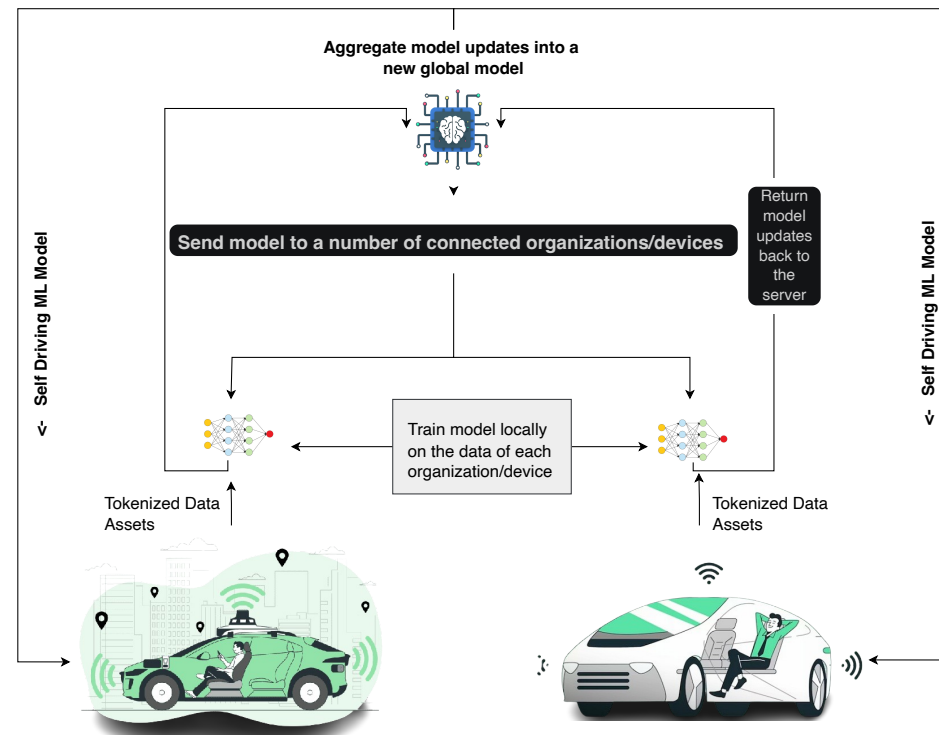


Figure 1. An illustration of the concept of distributed training for machine learning models in the context of autonomous vehicles.

Despite its advantages, FL faces challenges in ensuring data integrity and trustworthiness. Blockchain technology addresses these limitations by providing a decentralized, immutable ledger for recording data transactions and model updates [8,9]. By automating governance through smart contracts, blockchain ensures data authenticity, compliance, and secure access across IIoT assets [10–12]. These capabilities are vital for fostering trust in data-driven decision-making within industrial environments.

Equally important is the integration of XAI to overcome the “black-box” nature of AI models. XAI techniques enhance the interpretability of machine learning outputs, offering insights into decision-making processes. This transparency builds stakeholder trust and ensures compliance with regulatory standards, particularly in high-stakes industrial applications [13]. Combined with self-optimization capabilities, where systems autonomously adapt to changing conditions, XAI strengthens the utility of digital twins in dynamic IIoT settings [14].

This paper presents a holistic framework that synergizes FL, blockchain, and XAI to address security, privacy, and interpretability challenges in IIoT. The contributions of this work are as follows:

- Introducing blockchain for secure, transparent, and decentralized data management within IIoT ecosystems.
- Leveraging FL to enable privacy-preserving, decentralized training of machine learning models on edge devices.
- Employing XAI to enhance the interpretability of AI-driven decisions, fostering trust among stakeholders.

- Providing a scalable and resilient framework for operational efficiency and real-time decision-making in IIoT.

The integration of these technologies offers a transformative approach to managing digital twins, driving secure, efficient, and trustworthy operations in IIoT environments.

The rest of this paper is structured as follows: Section 2 reviews related work, examining how blockchain integration with digital twins has enhanced various sectors, such as construction and predictive maintenance. Section 3 details the methodology and overall framework, including data preprocessing, secure data management with blockchain, and FL with secure aggregation. Section 4 discusses the assessment of the proposed framework, focusing on simulation setups, learning performance, operational efficiency, and security implications. Finally, Section 5 concludes the paper, summarizing the findings and suggesting directions for future research.

2. Related Work

The integration of FL, blockchain, and digital twin technologies has garnered significant attention in IIoT due to their potential to enhance security, privacy, and operational efficiency. These technologies address critical challenges in decentralized industrial systems, including data integrity, secure communication, and scalable model training. However, gaps remain in fully leveraging their collective potential, particularly in ensuring transparency and interpretability through XAI. This section discusses the current state of research on FL, blockchain, and digital twins while highlighting research gaps that the proposed framework addresses.

FL enables decentralized training of machine learning models directly on edge devices, preserving privacy and minimizing bandwidth usage. Nguyen et al. [7] emphasized FL's application in data sharing, caching, and attack detection within IoT networks. Wu et al. [15] discussed blockchain-enabled FL frameworks that facilitate decentralized model training in secure environments. Yang et al. [16] introduced adaptive optimization techniques to ensure efficient FL performance under dynamic industrial conditions, while Jiang et al. [17] developed a resource-efficient FL model integrating blockchain with sharding, optimizing performance in digital twin-driven IIoT systems. Salim et al. [18] proposed an FL-based cyber threat intelligence (CTI) framework, showcasing FL's potential in securing IIoT data flows. Lv et al. [19] demonstrated a decentralized FL framework leveraging blockchain for enhanced data privacy and model security. Ramu et al. [20] extended FL applications to smart city digital twins, emphasizing its scalability and privacy-preserving capabilities. Despite these advances, challenges persist in adapting FL to the varying conditions and resource constraints inherent in IIoT environments.

Blockchain technology complements FL by providing a decentralized, tamper-proof ledger for data management [21]. Prathiba et al. [22] highlighted blockchain's role in strengthening FL and digital twin ecosystems, offering transparent and immutable data histories. Kumar et al. [23] underscored blockchain's utility in securing digital twin communications through deep learning models, enhancing data confidentiality and system resilience. Jiang et al. [24] introduced a cooperative FL framework where blockchain verifies model updates, ensuring accuracy in decentralized learning. Maurya et al. [25] demonstrated blockchain's potential in federated transfer learning, preserving data privacy and authentication in IIoT. Cai et al. [26] and Sasikumar et al. [27] proposed blockchain-based trust mechanisms for digital twins, emphasizing their critical role in secure data exchange and decentralized control.

Further studies such as Zhao et al. [28] introduced a blockchain-based system for secure sensing data processing and logging, ensuring data integrity and accessibility in distributed environments. Their work underscores the importance of blockchain in IIoT

applications where secure data management is paramount. Gangwani et al. [29] proposed a novel approach using DAG-based blockchain combined with masked authentication messaging protocols to secure environmental IoT data, demonstrating remarkable potential for improving efficiency and scalability in IoT-based systems. Although these studies demonstrate blockchain’s potential to enhance IIoT systems, scalability and computational efficiency have always been open challenges.

Digital twins, as virtual representations of physical assets, play a pivotal role in IIoT by enabling real-time monitoring, predictive maintenance, and operational optimization. Sadri et al. [30] proposed a blockchain- and AI-driven framework for predictive operation and maintenance, leveraging secure data storage and automation. Hunhevicz et al. [4] explored blockchain’s application in digital twins for performance-based smart contracts, enhancing transparency and accountability. Leng et al. [31] combined edge learning with blockchain to improve predictive maintenance, while Ramu et al. [20] highlighted digital twins’ potential in collaborative urban data analysis. Zhou et al. [32] applied federated modeling to smart grids, optimizing energy management in digital twin environments. Lv et al. [19] and Abdullah et al. [33] showcased blockchain-enabled digital twins’ ability to deliver low-latency, secure data-driven insights. Despite these advances, scalability and interoperability in diverse IIoT settings remain significant gaps.

XAI has gained prominence as a means of addressing the “black-box” nature of AI models in IIoT. Jan et al. [34] discussed the challenges of AI interpretability in Industry 4.0, advocating for transparent and regulatory-compliant AI solutions. Liu et al. [35] and Deng et al. [36] emphasized the importance of integrating XAI into FL and blockchain networks to foster trust and transparency in industrial applications. However, standardized methodologies for deploying XAI in dynamic IIoT environments are lacking, and existing solutions often fail to balance computational efficiency with interpretability.

As shown in Table 1, prior research primarily focuses on isolated applications of blockchain, FL, or digital twins. The proposed framework distinguishes itself by integrating these technologies with XAI, addressing the critical challenge of interpretability in high-stakes industrial environments. This technological synergy ensures that the framework not only enhances security and scalability but also provides transparency in its decision-making processes, a feature often overlooked in existing solutions. Moreover, the target applications of the reviewed works, as outlined in Table 1, are typically confined to specific domains, such as healthcare or smart cities. In contrast, the proposed framework demonstrates broader applicability, encompassing diverse IIoT scenarios. This versatility allows it to adapt to various industrial sectors, making it a more robust and comprehensive solution.

Table 1. Comparative analysis of related works.

Reference	Target Application	Technologies Employed	Key Contributions
Jiang et al. [17]	IIoT	FL, Blockchain	Developed resource-efficient FL with sharding, enhancing scalability and security for digital twin-driven IIoT.
Ramu et al. [20]	Smart Cities	FL, Digital Twins	Highlighted FL’s scalability and privacy-preserving capabilities for collaborative urban planning.
Huang et al. [37]	IoT Data Management	Blockchain, Consensus Mechanisms	Introduced a consensus mechanism for secure data exchange in IoT environments, emphasizing scalability and reliability.
Kaul et al. [38]	Healthcare (Cancer Care)	AI, Digital Twins	Leveraged digital twins and AI for predictive healthcare applications, focusing on patient-specific modeling.
Suhail et al. [39]	IIoT	Blockchain, Digital Twins	Surveyed trends and challenges of integrating blockchain and digital twins in IIoT applications.
Sadri et al. [30]	Predictive Maintenance in Construction	Blockchain, AI, Digital Twins	Proposed a framework combining blockchain and AI-driven digital twins for predictive operations in the built environment.
Our Framework	IIoT	Blockchain, FL, XAI	Integrates blockchain, FL, and XAI for secure, scalable, and interpretable digital twin management in IIoT environments.

In summary, despite advancements in FL, blockchain, and digital twins, research gaps still persist, particularly in their integration into a unified framework, addressing computational overhead in resource-constrained IIoT environments, and validating scalability through real-world implementations. This paper bridges these gaps by proposing a comprehensive framework that combines FL, blockchain, and XAI to enable secure, privacy-preserving, and interpretable digital twin operations, advancing scalable and trustworthy IIoT systems.

3. Framework Design and Deployment

To address the challenges identified in the related work, we propose a comprehensive framework that integrates blockchain technology, FL, and XAI to enhance the transparency, functionality, and security of digital twins in IIoT environments. This section details the methodologies employed, including data preprocessing, secure data management, decentralized model training, and the implementation of smart contracts, to achieve a robust and scalable system for optimizing digital twins.

3.1. Data and Data Preprocessing

To support the proposed framework effectively, robust data preparation and privacy-preserving mechanisms are critical, especially given the sensitivity of data in IIoT applications. The CIFAR-10 dataset [40] was selected due to its structured organization of diverse image classes, making it ideal for simulating real-world IIoT scenarios. While multiple datasets were considered during the conceptualization phase, CIFAR-10 was ultimately chosen as the sole dataset for experimental evaluation due to its suitability for benchmarking and manageable size, allowing efficient experimentation and analysis within the scope of this study.

To ensure robust data privacy and performance, preprocessing integrates Homomorphic Encryption (HE) to maintain data confidentiality during computations. HE enables secure operations directly on encrypted data, eliminating the risk of exposing sensitive information. Additionally, blockchain-based smart contracts enforce strict access control and data traceability, providing an immutable audit trail for all interactions. These mechanisms enhance the integrity and trustworthiness of the data pipeline, ensuring privacy while enabling accurate and secure FL model training. Specifically, a leveled HE scheme, such as CKKS or BFV [41], is implemented to support essential operations like addition and multiplication on encrypted sensor data. This approach enables manipulation of the data while ensuring the raw information remains inaccessible to unauthorized parties. Each data point x from sensors is encrypted into ciphertext $E(x)$ via the encryption function $E(x) = \text{Encrypt}(x, pk)$, where pk denotes the public key for encryption. This ciphertext is then used in all subsequent computations, which preserves the data's privacy as it flows through the FL model.

To enhance data security and streamline management, we integrate blockchain-based smart contracts within the framework. These contracts manage data access and ensure only authorized entities can access and process the encrypted data, safeguarding sensitive information at every stage. Core functionalities of the smart contract include access control, whereby only users with valid cryptographic keys can retrieve or interact with the encrypted data, maintaining secure access across stakeholders. The contract also provides data provenance and auditing functionalities by logging every interaction, creating a transparent record of all data operations. This level of traceability strengthens data integrity and fosters trust within the collaborative learning environment. Moreover, for applications involving multiple stakeholders (such as in an FL setup), the smart contract automates data

sharing agreements, ensuring that data exchange is controlled and unauthorized access is prevented.

Once encrypted, data undergoes processing for machine learning model training. The HE scheme enables privacy-preserving computations, allowing machine learning algorithms to operate directly on encrypted data without decryption. For each encrypted data point $E(x)$, mathematical operations, such as summation and multiplication, can be applied to the ciphertext itself. For instance, the encrypted sum of data points $E(y) = E(x_1) + E(x_2)$ can be computed directly from the encrypted inputs $E(x_1)$ and $E(x_2)$. This capability is extended to more complex operations, such as gradient descent, which is essential for model optimization in machine learning. By performing these operations on encrypted data, the framework preserves the privacy of sensor data throughout the training process. Furthermore, after training, critical model outputs, such as the model weights, remain encrypted and can only be decrypted by authorized parties with access to the private decryption key.

After the machine learning model completes training, authorized personnel decrypt the encrypted results to evaluate model performance. This decryption process involves using a private key sk , where $x = \text{Decrypt}(E(x), sk)$ reveals the encrypted model weights or performance metrics to verified users only. This decryption step is exclusively performed by entities with explicit authorization, preserving the confidentiality of the data throughout the FL pipeline. By incorporating Homomorphic Encryption and blockchain-based smart contracts, the data preprocessing pipeline ensures that privacy, security, and data integrity are maintained. At the same time, these privacy-preserving mechanisms enable sophisticated machine learning analysis on sensitive data, contributing to a secure and resilient framework in IIoT applications.

3.2. Blockchain for Secure Data Management

Blockchain technology provides a secure and transparent mechanism for managing data access and updates in FL environments, as depicted in Figure 2. Its implementation in the proposed framework relies on two key components:

- **Smart contracts:** Smart contracts are utilized to govern data transactions and access controls. These contracts enforce predefined rules, ensuring data traceability and auditability. The state of a smart contract represents its current condition, while transactions dictate changes to this state. When a transaction meets the specified conditions of the contract, it transitions to a new valid state, maintaining the integrity and compliance of data governance processes. This ensures secure and transparent data management, as illustrated in Figure 2.
- **Digital twin integration:** Each physical asset in the system is represented by a corresponding digital twin that simulates its real-time status. The relationship $T_i = f(D_i)$ expresses that real-time data from an IoT device (D_i) updates its associated digital twin (T_i). Here, T_i represents the digital twin of the i^{th} physical asset, and D_i represents the data collected from the i^{th} IoT device. This ensures that the digital twin accurately reflects the current state of its physical counterpart, facilitating real-time monitoring and management.

Blockchain is a critical component of the proposed framework, addressing key challenges in FL and IIoT environments. Its integration ensures secure and decentralized data management through an immutable ledger that records data transactions and model updates, using Merkle Trees to verify data integrity efficiently. Smart contracts play a pivotal role by validating model updates with cryptographic signatures, ensuring that only authenticated updates are integrated into the global model. Additionally, blockchain enhances privacy by enforcing role-based access control through smart contracts, in con-

junction with HE, to maintain the confidentiality of sensitive data. Lightweight consensus mechanisms, such as Delegated Proof of Stake (DPoS), are employed to balance security and computational efficiency, making the implementation scalable for resource-constrained IIoT environments. Moreover, blockchain's inherent transparency provides a verifiable audit trail of all operations, meeting regulatory compliance requirements and enhancing trust among stakeholders. These features collectively validate the inclusion of blockchain as an indispensable element of the framework, directly addressing security, scalability, and privacy challenges.

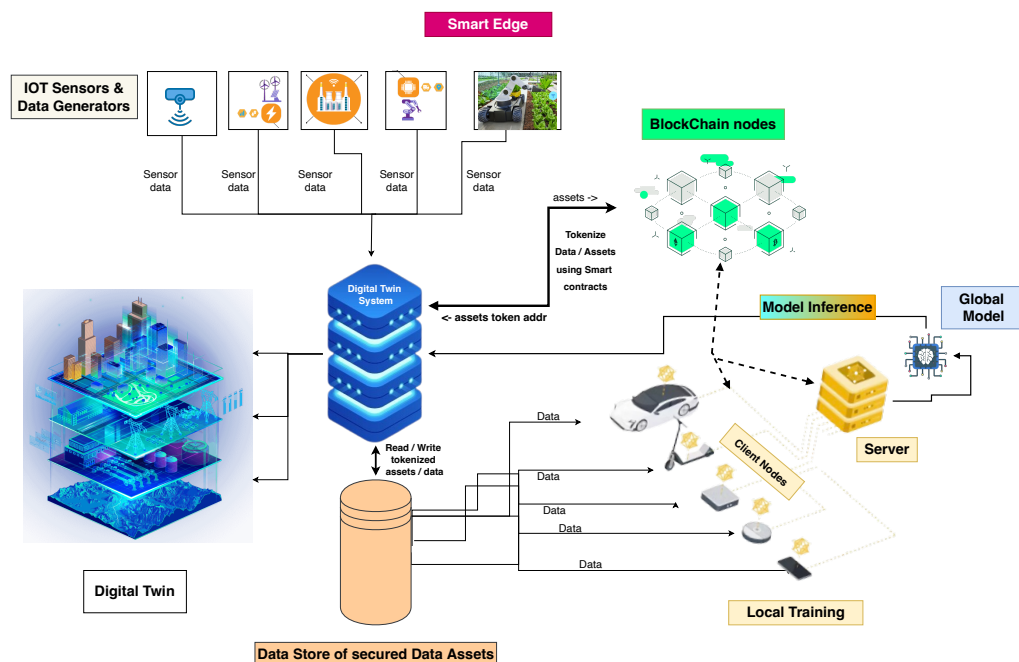


Figure 2. Workflow of the proposed FL framework, showcasing how anonymized data from IoT sensors are processed using XAI and blockchain technology for secure and collaborative global model training and deployment within a digital twin system.

3.3. FL with Secure Aggregation

FL enables collaborative model training while keeping data distributed on edge devices such that $w_{t+1} = w_t + \eta \sum_{k=1}^K \Delta w_t^k$, where w_t , Δw_t^k , and η represent the model weights at iteration t , updates from the k^{th} device, and the learning rate, respectively. Here, the local model updates are aggregated from multiple devices to create a new global model (w_{t+1}).

3.4. Proposed Framework

Figures 2 and 3 collectively illustrate the key components, interactions, and workflow of the proposed FL framework, emphasizing its integration of blockchain technology, XAI, and digital twin systems in IIoT environments. These figures provide a visual representation of how the framework addresses critical challenges, such as data privacy, security, and scalability.

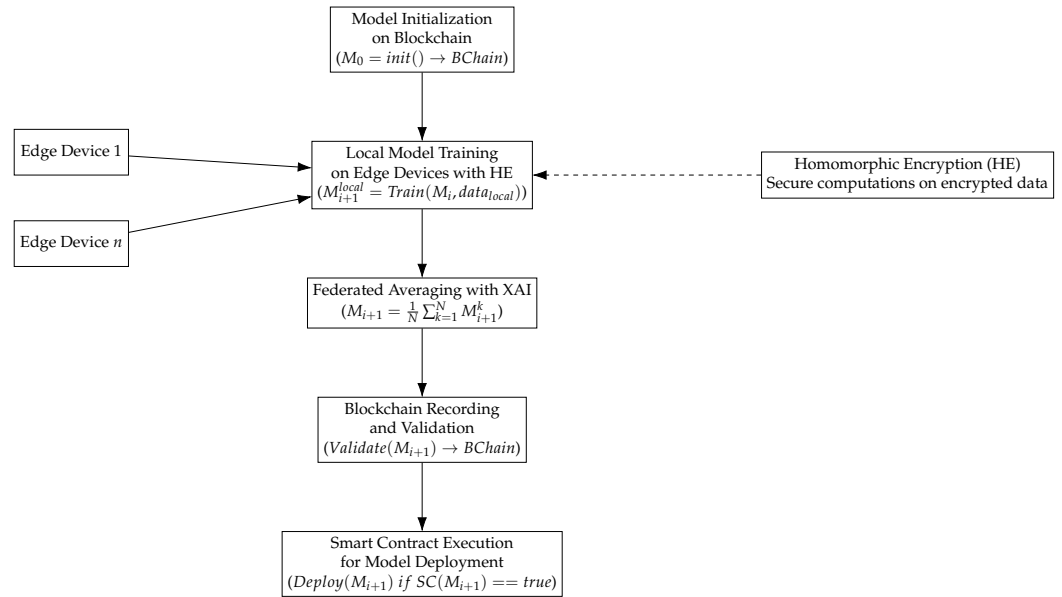


Figure 3. Workflow of the proposed framework for blockchain-assisted FL with digital twins in IIoT, incorporating HE for privacy-preserving local training and blockchain for secure model validation and deployment.

Figure 2 highlights the high-level architecture, showing how anonymized data from IoT sensors are processed at the edge devices using privacy-preserving techniques like HE. The framework performs local model training on encrypted data, ensuring that the raw data remain secure throughout the process. These locally trained models are aggregated into a global model through the FL process, with blockchain ensuring the integrity and authenticity of the updates. The figure further illustrates the use of blockchain as a decentralized and immutable ledger for recording model updates and facilitating secure collaboration between edge devices and digital twins. The integration of XAI enhances the interpretability of the global model, fostering trust and usability in industrial settings.

Building on this, Figure 3 provides a detailed workflow of the framework, presenting a step-by-step view of its processes. The workflow begins with model initialization on the blockchain, where the initial model parameters are securely recorded to ensure transparency and integrity. Next, local model training occurs on edge devices using HE, allowing computations to be performed on encrypted data. The updates from edge devices are then aggregated using federated averaging, with XAI integrated to analyze feature importance and refine the global model. The aggregated global model is validated and recorded on the blockchain in the blockchain recording and validation stage, ensuring only authenticated updates are included. Finally, smart contract execution automates the deployment of the validated model to edge devices if predefined quality standards are met.

Together, these figures convey the multi-layered approach of the proposed framework, demonstrating how blockchain, FL, and XAI converge to enable secure, scalable, and privacy-preserving model training and decision-making processes. By providing both a high-level architectural view (Figure 2) and a detailed workflow (Figure 3), the figures ensure clarity and relevance, bridging the conceptual design and practical implementation of the framework.

3.4.1. Model Initialization on Blockchain

The model initialization process establishes the baseline for collaborative learning by creating and storing initial machine learning model parameters, denoted as M_0 , on the blockchain. Specifically, this step initiates the model as $M_0 = \text{init}() \rightarrow BChain$, where the blockchain (referred to as $BChain$) securely records the model's parameters. By record-

ing the model on the blockchain from the outset, the framework ensures an immutable and transparent history of the model's evolution, facilitating security and trust in the collaborative learning process.

This initialization serves as a foundation for training across a federated network of edge devices. By establishing M_0 on the blockchain, the framework provides a verified starting point that enhances accountability and prevents unauthorized modifications, aligning with the principles of decentralized trust and data integrity.

3.4.2. Local Model Training on Edge Devices

In the next phase, each edge device engages in local model training using its own data to update the model. This process is defined as $M_{i+1}^{local} = \text{Train}(M_i, \text{data}_{local})$, where M_i represents the global model parameters at iteration i , and data_{local} refers to the data held locally on each device. This design preserves user privacy by eliminating the need for raw data exchange, allowing devices to train the model independently.

Local training enables each edge device to leverage unique data characteristics relevant to its environment, enhancing model adaptability. By performing this step locally, the framework benefits from diverse data sources without compromising privacy, reinforcing FL's core objective of decentralized model training with data security.

3.4.3. Federated Averaging with XAI Integration

The aggregation equation simplifies the federated averaging process for clarity, but it does not merely involve a raw summation of model parameters. Instead, it incorporates weighting factors based on hyperparameters such as the number of data samples on each participating device. Specifically, the weights assigned to each local model M_{i+1}^k are proportional to the size of the local dataset, ensuring that devices contributing more significant data influence the global model proportionally. The adopted aggregation equation can be expressed as:

$$M_{i+1} = \sum_{k=1}^N \frac{n_k}{n} M_{i+1}^k, \quad (1)$$

where n_k is the number of data samples on device k , and $n = \sum_{k=1}^N n_k$ represents the total number of samples across all devices. This weighted approach prevents devices with smaller datasets from disproportionately impacting the global model and enhances the robustness and generalizability of the aggregated model. The implementation also considers potential adjustments to the hyperparameters, such as learning rates or local training epochs, to fine-tune the contributions of each device further. A discussion of these nuances highlights the thoughtfulness of the aggregation strategy and addresses the concern raised about oversimplification.

The weights $\frac{n_k}{n}$ in Equation (1) are determined based on the number of data samples n_k on each device, where $n = \sum_{k=1}^N n_k$. This ensures that devices with larger datasets have a proportionally greater influence on the global model, reflecting their statistical significance. Additional factors can influence the weights, such as data quality (e.g., reliability and completeness), model convergence (e.g., lower loss or higher accuracy), and device reliability (e.g., historical consistency). A generalized weighting function w_k can be formulated as:

$$w_k = \alpha \cdot \frac{n_k}{n} + \beta \cdot q_k + \gamma \cdot r_k, \quad (2)$$

where q_k represents data quality, r_k reflects reliability, and α, β, γ are hyperparameters. This flexible approach allows the aggregation process to incorporate both the quantity and quality of contributions, ensuring fairness and robustness in the FL framework.

To enhance this aggregation, the framework integrates XAI techniques post-aggregation, generating explanations (e.g., attention maps and gradients) to identify critical features and assess the importance of various data inputs. This insight-driven approach allows the system to refine model aggregation further and prioritize high-impact data as follows:

- By incorporating XAI, the framework adjusts the FL process to optimize the impact of significant features. XAI methods identify feature importance and assist in prioritizing adjustments, particularly for sensor inputs prone to errors. This adjustment mechanism aims to ensure that the model performs robustly across a variety of operational conditions, increasing interpretability and reliability.
- To reinforce the XAI-driven insights, each local model's contribution is weighted according to its identified importance. Let W_{i+1}^k represent the weight assigned to the model update from device k after update cycle $i + 1$. The revised aggregation process is represented as:

$$M_{i+1} = \frac{\sum_{k=1}^N W_{i+1}^k M_{i+1}^k}{\sum_{k=1}^N W_{i+1}^k} \quad (3)$$

This approach improves the global model by emphasizing data sources that contribute to more accurate and resilient predictions, guided by XAI insights.

- During local training on each edge device, XAI further customizes the training process by prioritizing data points associated with historical model errors. The local loss function $L^k(M^k, x_t^k)$ is modified using an importance score α_t^k :

$$L_{XAI}^k(M^k, x_t^k) = \alpha_t^k \cdot L^k(M^k, x_t^k) \quad (4)$$

This prioritization helps the model adapt to challenging data points, fostering a more robust learning process and reducing the impact of recurrent errors.

- To address edge cases, the framework adopts a prototype and criticism approach. The loss function emphasizes data that XAI identifies as either prototypical or problematic, allowing the model to focus on these cases. This weighted function is given by:

$$L_{proto-crit}^k = \beta L^k(M^k, x_{proto}) + \gamma L^k(M^k, x_{crit}) \quad (5)$$

Here, β and γ prioritize training on essential data, enhancing the model's ability to manage edge cases effectively.

- The framework's integration of XAI establishes a self-optimizing feedback loop that continuously refines the model based on real-time insights. This iterative process, represented as:

$$M_{i+1} = \frac{\sum_{k=1}^N W_{i+1}^k (M_{i+1}^k + \Delta M_{XAI}^k)}{\sum_{k=1}^N W_{i+1}^k}, \quad (6)$$

incorporates XAI-based adjustments, making the model more resilient and adaptable to dynamic conditions in IIoT environments.

3.4.4. Blockchain Validation and Smart Contract Deployment

Once the global model is updated, it undergoes validation to ensure it meets predefined accuracy thresholds. This process guarantees that only high-quality model updates are considered for deployment. The validated model M_{i+1} is recorded on the blockchain as $Validate(M_{i+1}) \rightarrow BChain$, creating an immutable and transparent record of each model version. By recording validated updates only, the framework prevents unauthorized modifications and ensures data integrity, providing a robust mechanism to maintain trust in the FL process.

Following validation, the framework employs smart contracts to evaluate the updated model's quality before deployment. These contracts enforce predefined conditions, expressed as $Deploy(M_{i+1}) \text{ if } SC(M_{i+1}) == true$, ensuring that only models meeting specific performance standards are deployed. This automated compliance check safeguards model reliability, offering an additional security layer critical for real-world applications. Smart contracts not only streamline the deployment process but also enhance the overall trustworthiness of the system by automating governance and reducing the potential for human error.

Algorithm 1 outlines the overarching workflow of the proposed blockchain-assisted FL framework. It integrates distinct yet interconnected procedures, including model initialization, local training on edge devices, federated model aggregation, blockchain validation, and smart contract deployment. Each step operates at a different layer of the system, collectively contributing to the framework's end-to-end functionality. This modular design ensures scalability, flexibility, and maintainability, allowing components to be adapted or enhanced independently while preserving the overall system integrity. By clearly delineating these tasks, the algorithm demonstrates how the framework achieves its goals of enhancing security, privacy, and operational efficiency in IIoT applications.

Algorithm 1 Blockchain-assisted FL framework with HE and digital twins in IIoT (overall procedure).

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1: Input: Initial model  $M_0$ , edge devices data  $\{data_{local}^k\}_{k=1}^N$ , blockchain  $BChain$ , encryption key  $K$ 
2: Output: Trained global model  $M_{final}$  securely recorded on blockchain
3: procedure INITIALIZE MODEL ON BLOCKCHAIN
4:    $M_0 = init()$ 
5:    $BChain \leftarrow Hash(M_0)$  ▷ Record hash of initial model on blockchain
6: end procedure
7: procedure LOCAL TRAINING WITH HOMOMORPHIC ENCRYPTION
8:   for each edge device  $k = 1, \dots, N$  do
9:      $Enc(data_{local}^k) = Encrypt(data_{local}^k, K)$  ▷ Encrypt local data using HE
10:     $Enc(M_{i+1}^{local}) = Train(Enc(M_i), Enc(data_{local}^k))$  ▷ Train locally on encrypted data
11:   end for
12: end procedure
13: procedure FEDERATED AVERAGING WITH XAI INTEGRATION
14:   Collect  $Enc(M_{i+1}^{local})$  from all edge devices
15:    $Enc(M_{i+1}) = Aggregate(\{Enc(M_{i+1}^{local})\}_{k=1}^N)$  ▷ Aggregate encrypted models using weighted averaging
16:    $M_{i+1} = Decrypt(Enc(M_{i+1}), K)$  ▷ Decrypt aggregated model
17:   for each prediction in  $M_{i+1}$  do
18:     Generate XAI explanations (e.g., feature importance, attention maps)
19:   end for
20: end procedure
21: procedure BLOCKCHAIN VALIDATION AND RECORDING
22:   if Validate( $M_{i+1}$ ) then
23:      $BChain \leftarrow Hash(M_{i+1})$  ▷ Record hash of validated model on blockchain
24:   end if
25: end procedure
26: procedure SMART CONTRACT EXECUTION AND DEPLOYMENT
27:   if SmartContract( $M_{i+1}$ ) == true then
28:     Deploy  $M_{i+1}$  to edge devices ▷ Deploy global model if conditions are met
29:   end if
30: end procedure
31: return  $M_{final}$ 

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4. Evaluation

This section evaluates the proposed blockchain-assisted FL framework for self-optimizing digital twins in IIoT environments. It focuses on assessing the performance, scalability, and security contributions of the integrated technologies—FL, blockchain, and XAI—and demonstrates how these components collectively enhance the framework’s effectiveness in real-world scenarios.

To provide a clearer understanding of the framework’s workflow, Figure 4 illustrates the preprocessing stages and the key elements of the proposed framework, providing a clear depiction of the FL process. The diagram begins with data collection at edge devices, followed by local data preprocessing stages, including cleaning and normalization, ensuring data quality. The workflow then transitions to secure local model training with homomorphic encryption, which preserves privacy. Federated aggregation, facilitated by blockchain, ensures secure and immutable communication, while XAI enhances interpretability by providing insights into model behavior. The refined model is subsequently deployed to edge devices via smart contracts, completing the process. This comprehensive visualization enhances understanding of the framework’s workflow and its practical application in IIoT environments.

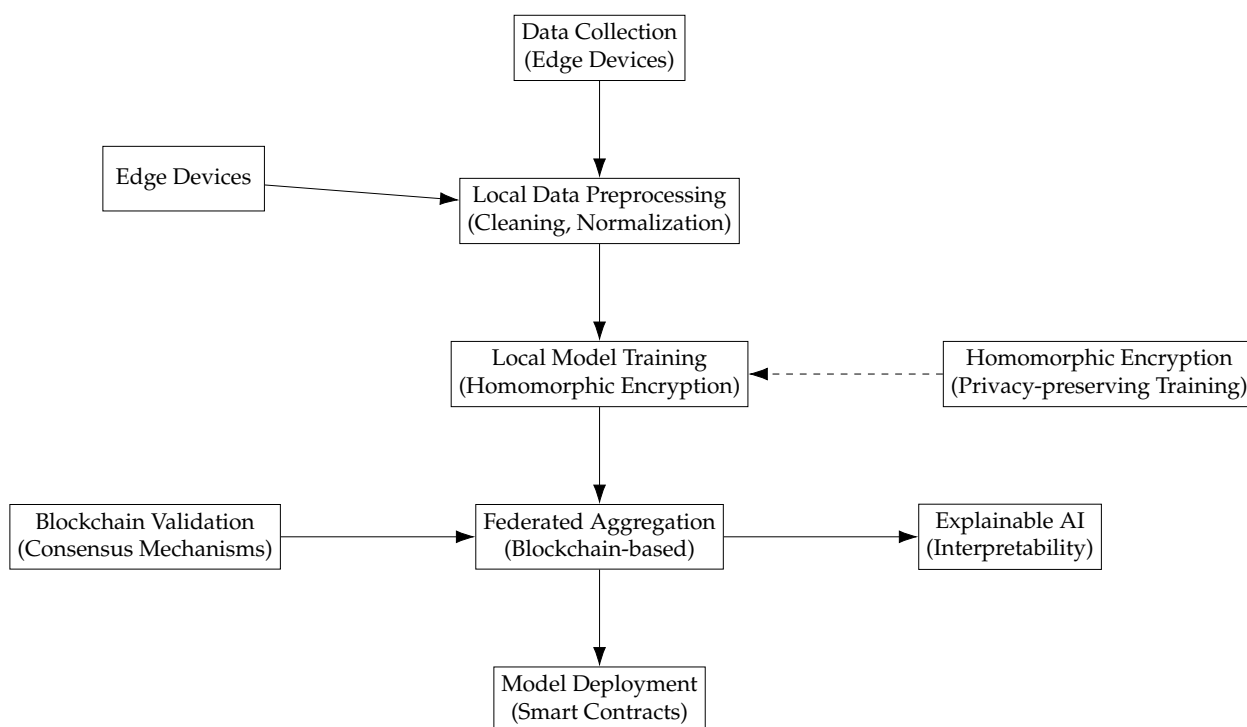


Figure 4. Workflow of the proposed framework for blockchain-assisted FL with digital twins in IIoT, depicting preprocessing stages, FL, blockchain validation, and XAI integration for interpretability.

4.1. Experiment Setup

4.1.1. Testbed and Resources

The FL simulation was conducted using two client nodes over three training rounds with the CIFAR-10 dataset [42]. Each client node operated independently, processing data locally and updating model parameters without transmitting sensitive information. This decentralized approach ensures data privacy and security by eliminating the need for centralized data storage, reducing the risk of cyberattacks. The experiments utilized a Tesla T4 GPU, featuring a 1.59 GHz clock rate, 40 cores, and 16 GB of memory with a maximum bandwidth of 300 GB/s. The GPU’s computational capabilities include 8.1 TFLOPS for FP32 and 65 TFLOPS for FP16 operations, providing strong performance for machine learning

tasks. These resources and configurations supported efficient and privacy-preserving FL simulations.

The experimental setup aligns with the proposed blockchain-assisted FL framework's objectives by simulating a distributed IIoT environment. Using two client nodes and three training rounds ensures a balance between computational efficiency and model convergence while preserving data privacy. The CIFAR-10 dataset [42] emulates the heterogeneous data distributions typical in IIoT scenarios. The Tesla T4 GPU, with its high memory bandwidth (300 GB/s) and optimized FP32/FP16 performance, supports efficient and scalable FL simulations. This configuration demonstrates the framework's feasibility for privacy-preserving and resource-efficient IIoT applications.

4.1.2. Frameworks Utilized

To support secure, decentralized, and efficient training and management of AI models in the proposed framework, the following frameworks were employed:

1. Web3: Provides tools for interacting with the Ethereum blockchain, enabling secure data transactions, smart contract execution, and access control. It ensures secure and transparent data management through immutable record-keeping (an intrinsic feature of blockchain technology).
2. Flower framework: An open-source framework for FL. It allows for decentralized model training on edge devices. This maintains data privacy and security by keeping data local and sharing only model updates.
3. Pytorch: A flexible, open-source machine learning library for building and training neural network models. It supports complex AI model development, essential for our proposed framework.

4.2. Key Results

This subsection highlights the framework's predictive accuracy, training efficiency, and scalability in IIoT scenarios, showcasing its ability to address critical industrial challenges through FL and blockchain integration.

4.2.1. Model Performance

The framework achieved a testing accuracy of 95%, demonstrating robust predictive capabilities in IIoT scenarios. The federated approach allowed the model to leverage diverse local data distributions across client nodes, contributing to improved generalization. Training logs showed a consistent reduction in loss, culminating in a final training loss of 0.0150 and an accuracy of 0.9972, as shown in Figure 5. These metrics validate the framework's suitability for predictive maintenance and other real-world applications.

In this experiment, FL enabled the aggregation of locally trained models, ensuring that the global model benefited from diverse and distributed data while preserving privacy. The use of weighted federated averaging further optimized contributions based on node-specific data quality and volume, improving overall performance.

4.2.2. Training Efficiency

The average training time per epoch was 294.48 s, with a total training duration of 427.12 s for 50 epochs. Data processing and model update latency averaged 300 milliseconds, which is critical for real-time IIoT applications.

The blockchain in this experiment facilitated secure model updates by ensuring integrity and authenticity of aggregated model parameters. Its lightweight consensus mechanism, DPoS, minimized computational overhead, contributing to efficient training.

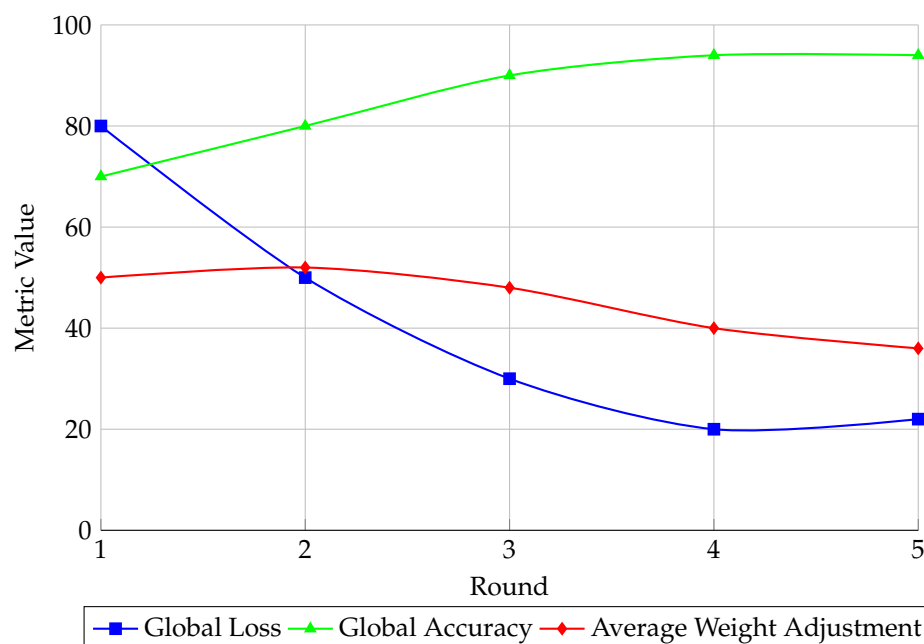


Figure 5. Global loss over rounds (top subfigure), global accuracy over rounds (middle subfigure), and average weight adjustment over rounds (bottom subfigure).

4.2.3. Scalability

In this experiment, the proposed framework demonstrated robust scalability as follows:

- **Device count:** Performance remained stable with up to 1000 edge devices. Training time increased slightly to 84.40 s per epoch, while latency rose to 350 milliseconds.
- **Data volume:** The framework efficiently handled varying data sizes from 10 GB to 100 GB, maintaining accuracy and efficiency.

From the blockchain impact on scalability perspective, the blockchain's decentralized ledger allowed seamless integration of additional devices without compromising security or increasing risks of bottlenecks.

4.3. Discussion

The following discussion provides a comprehensive analysis of the experimental results, highlighting the contributions of the proposed blockchain-assisted FL framework to IIoT applications. Key aspects such as learning performance, operational efficiency, security, and model interpretability are examined to demonstrate the framework's effectiveness in addressing privacy, scalability, and transparency challenges in dynamic industrial environments.

4.3.1. Learning Performance

The results projected in Figure 5 demonstrate the consistent reduction in global loss across training rounds, coupled with an increase in global accuracy. These trends confirm that the model successfully generalized across distributed datasets, enhancing predictive accuracy. The average weight adjustments depicted in Figure 4 further validate the effectiveness of the FL aggregation mechanism, showing convergence toward an optimal global model by leveraging diverse local data contributions.

To further analyze the role of FL in the experimental results, Figure 6 highlights the contributions of individual client nodes to the global model's training process, demonstrating the diversity of local datasets and training conditions. FL's weighted aggregation mechanism effectively balances these variations, enabling the global model to benefit from heterogeneous data while maintaining privacy. Nodes with higher data quality and larger

datasets had a more significant influence on the global model, reflecting the robustness of the FL approach. This decentralized mechanism ensured consistent reduction in global loss and improved accuracy, as shown in Figure 5, while preserving data privacy. FL's ability to aggregate insights from distributed nodes without sharing raw data underscores its suitability for privacy-sensitive IIoT applications, making it a pivotal component in achieving the experimental outcomes.

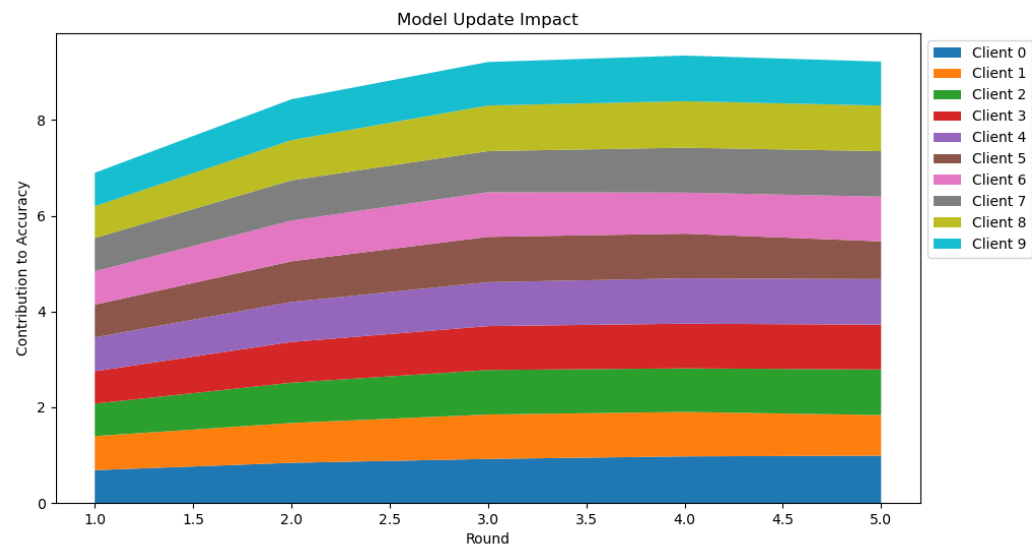


Figure 6. Model's update impact from each client node.

Additionally, based on Figure 6, the variations in contributions reflect the diversity of local datasets and training conditions, emphasizing the importance of FL in balancing these differences to create a robust global model. This figure also supports the claim that FL enables collaborative learning while preserving the uniqueness of each node's data, enhancing overall model performance.

These results demonstrated the framework's ability to generalize across diverse data distributions, as evidenced by the consistent reduction in loss and high accuracy. FL played a pivotal role by enabling collaborative training across nodes while maintaining data locality, which is crucial for privacy-sensitive IIoT environments.

4.3.2. Operational Efficiency and Security

The claim of reduced operational downtime in our framework is supported by the predictive maintenance capabilities of the proposed framework, which leverages FL to enhance model accuracy (95%) and generalization. High accuracy ensures timely detection of anomalies and faults, enabling preemptive interventions before failures occur. The integration of blockchain-based smart contracts automates the execution of maintenance tasks, reducing manual intervention delays. While specific downtime metrics are not measured in this study, the framework's architecture inherently supports more proactive and efficient operations by enabling real-time detection and response. This is consistent with the documented advantages of predictive maintenance systems in reducing unplanned interruptions in industrial environments.

From the security perspective, blockchain's decentralized architecture and immutable ledger ensured robust protection against unauthorized access and tampering. This feature is especially critical in maintaining trust among stakeholders in industrial environments.

4.3.3. Model Interpretability

To ensure that the AI models integrated within our blockchain-assisted FL framework for digital twins in IIoT are not only secure and efficient but also transparent and interpretable, advanced model interpretability techniques were incorporated. Understanding the decision-making process of AI models is critical for building trust and enhancing usability in real-world industrial applications.

To achieve this transparency, a continuous feedback loop using Local Interpretable Model-agnostic Explanations (LIME) was employed. LIME generated explanations for individual predictions made by the model, allowing stakeholders to assess the rationale behind each decision. The clarity and usefulness of these explanations were systematically evaluated, ensuring alignment with human intuition and domain knowledge. For instance, as shown in Figure 7, the model significantly relied on specific, high-impact image regions during prediction, providing evidence of its focus on relevant features.

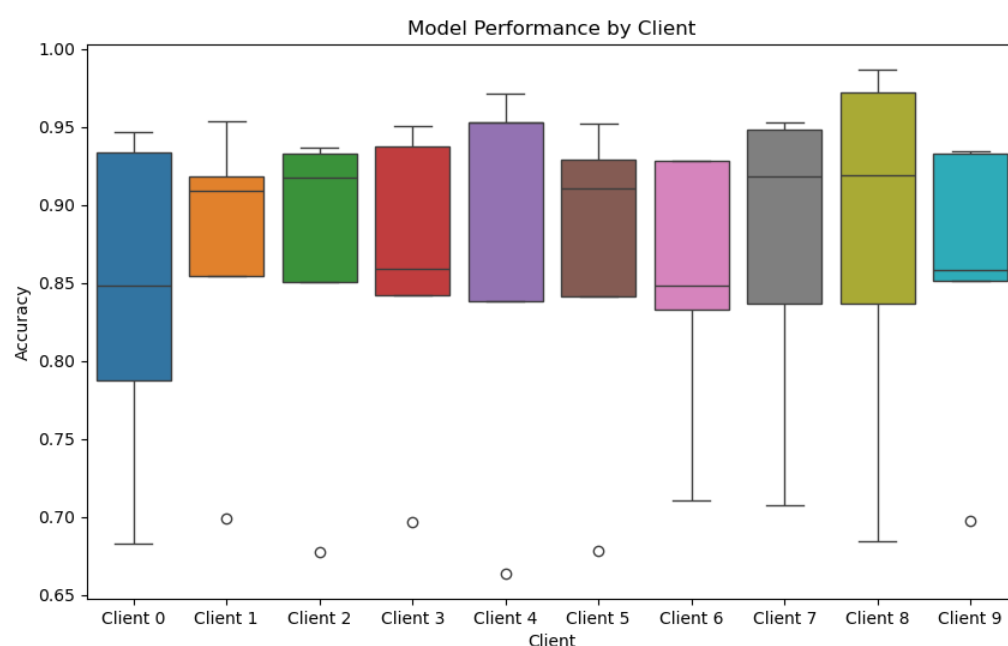


Figure 7. Model's performance contribution by each client node.

Building on this interpretability, the model became more adaptable and capable of real-time optimization by leveraging XAI-generated insights alongside raw data. These insights allowed the identification and prioritization of critical features, enhancing model accuracy and robustness. Moreover, the interpretability framework contributed to a self-optimizing digital twin system, designed specifically for dynamic IIoT environments.

XAI further aids the training process of the global model by facilitating feature selection and prioritization. Techniques like LIME analyze feature importance across diverse datasets from edge devices, identifying impactful features to reduce dimensionality and improve training efficiency. XAI also supports error analysis, refining local training by focusing on misclassified data or features associated with higher prediction errors. Additionally, XAI-generated feature importance scores influence the federated averaging process by assigning greater weight to models trained on critical features, ensuring the aggregated global model captures essential insights. This feedback loop enables the global model to dynamically adapt to changes in data distribution, improving robustness, generalization, and overall performance.

The resulting system is an interpretable, transparent, and effective FL model that not only strengthens operational effectiveness across industrial applications but also boosts

confidence in predictions and judgments. This adaptability ensures that the model evolves continually, aligning with the dynamic processes and diverse contexts inherent in IIoT operations. The integration of interpretability techniques thus serves as a cornerstone for building trust and achieving actionable insights in industrial systems.

XAI enhancements are summarized as follows:

- Enabled fine-grained analysis of model outputs, improving trust among users.
- Provided actionable insights into feature importance, guiding model optimization efforts.
- Established a feedback loop for self-optimization by prioritizing data associated with high-impact features or errors.

4.3.4. Practical Considerations for FL

The proposed framework ensures privacy, efficiency, and security in FL by addressing critical aspects of encryption, computation, and communication. HE enables secure computations on encrypted gradients, preserving data confidentiality during local training on edge devices. This ensures that sensitive information is never exposed, even in a distributed environment. While HE introduces computational overhead, the use of a Tesla T4 GPU with optimized FP32 and FP16 performance mitigates this impact, maintaining a balance between privacy and training efficiency.

To optimize communication, federated averaging with weighted contributions reduces the frequency and size of encrypted model updates transmitted across the network. Blockchain technology complements this process by securely logging aggregated updates, ensuring tamper-proof validation without increasing communication overhead. Together, these mechanisms demonstrate how the proposed framework achieves robust, privacy-preserving training suitable for dynamic IIoT environments.

4.3.5. Blockchain-Based Storage, Validation, and Consensus

The proposed framework ensures efficient and secure storage and validation of the global model using blockchain technology. Instead of storing the entire model, only a hash of the global model is recorded on the blockchain to ensure integrity while minimizing storage requirements. For instance, in our experiments with CIFAR-10, the model size was approximately 2 MB, and the hash sufficiently captured its integrity. Blockchain transactions, including model updates, typically take about 15 s to be recorded, leveraging Ethereum's average block time. Validation of these transactions is conducted by decentralized blockchain nodes, which use cryptographic signatures to ensure tamper-proof updates. A lightweight DPoS consensus mechanism further optimizes computational efficiency, balancing security and scalability for IIoT applications. This design ensures the framework remains robust, secure, and practical for real-world deployment.

4.3.6. Integration of HE and XAI

In the proposed framework, HE using the CKKS scheme ensures the security and privacy of local data and gradients during training on edge devices. The encrypted local models are transmitted to the central server, where aggregation is performed using homomorphic operations, eliminating the need for decryption during this process. Once the global model is aggregated, it is decrypted at the server end for further processing. This decrypted global model is then subjected to XAI techniques, such as feature importance analysis or interpretability mapping, to extract insights that enhance model transparency and guide future local training iterations. This integration of HE and XAI ensures that privacy is maintained at the edge level while leveraging interpretability to refine the global model and improve overall performance.

4.3.7. Framework Applicability to IIoT Environments

The proposed blockchain-assisted FL framework offers a robust solution to critical challenges faced in IIoT environments, even though its evaluation was conducted using a general-purpose dataset. The integration of FL minimizes reliance on centralized data processing, significantly reducing latency while ensuring privacy. Blockchain technology strengthens the framework by securing the aggregation process and providing immutable validation of model updates, thereby addressing common bottlenecks in distributed IIoT systems. These features position the framework as an efficient and scalable solution, particularly suited for dynamic and resource-constrained IIoT infrastructures.

Beyond its foundational evaluation, the framework's capabilities align closely with real-world IIoT requirements. For instance, the emphasis on XAI ensures interpretability in applications like predictive maintenance, where transparency in decision-making is crucial for operational safety and efficiency. Additionally, the framework's ability to handle heterogeneous data generated by various IIoT devices underscores its adaptability to complex industrial environments. By incorporating these design principles, the framework demonstrates clear potential for deployment in applications such as industrial wireless sensor networks (WSNs), where scalability, security, and adaptability are paramount.

4.4. Future Directions

In future work, the framework will explore mechanisms to address the impact of compromised or malfunctioning edge devices on global model integrity. Potential directions include implementing anomaly detection techniques to identify and exclude malicious updates during the aggregation process and developing adaptive weighting mechanisms that prioritize contributions from reliable nodes based on historical performance and data quality. Additionally, integrating blockchain-based trust scores for edge devices could further enhance the resilience of the FL process by ensuring only verified and trustworthy updates influence the global model. These enhancements will strengthen the robustness of the framework in real-world deployments.

To enhance the robustness of the FL model against adversarial attacks and data poisoning, future work will focus on integrating advanced mitigation strategies. These include implementing differential privacy to limit the influence of malicious updates, utilizing robust aggregation techniques, such as Krum or Trimmed Mean, to identify and exclude poisoned data contributions, and leveraging blockchain-based validation mechanisms to authenticate model updates. These enhancements aim to strengthen the security and reliability of the FL process, ensuring resilience in decentralized and potentially adversarial environments.

5. Conclusions

The proposed framework combines blockchain and federated learning (FL), with plans to integrate explainable AI (XAI), to deliver a secure, scalable, and interpretable solution for managing and optimizing digital twins in Industrial IoT (IIoT). This approach addresses challenges of privacy, security, and scalability, offering potential to transform industrial operations. Future work will increase the node count in simulations to better assess scalability under larger, realistic conditions, evaluating the impact of diverse datasets on efficiency and accuracy. We also plan to introduce incentive mechanisms for high-performing clients and exclude underperforming ones, focusing resources on meaningful contributions. These improvements will enhance system robustness, making it more suited for real-world applications, leading to secure and efficient industrial operations.

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